Keunhyoung Park

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Research Interest and Specialty

Machine Learning Interval Analysis, Novelty Detection, Additive Manufacturing of Concrete, Structural Health Monitoring, Design Optimization, Computational Structural Analysis and Design, Seismic Design

Education

Georgia Institute of Technology – Atlanta, GA — Aug 2020 – Present
Ph.D. in Civil Engineering

Pennsylvania State University – University Park, PA — Aug 2017 – Aug 2020
M.S. in Architectural Engineering, 2020

Yonsei University – Seoul, South Korea — Mar 2012 – Feb 2014
M.S. in Architectural Engineering, 2014

Yonsei University – Seoul, South Korea — Mar 2006 – Feb 2012
B.S. in Architectural Engineering, 2012

Research Activity

Graduate Student — Aug 2022 – Present

- Development of damage detection for structural health monitoring based on artificial neural network
  - Abnormal detection algorithm for physics-oriented data

Graduate Research Assistant (Penn State additive manufacturing team) — Oct 2017 – Jan 2020

- Development of additive manufacturing system for building a habitat structure
  - Parametric study of the extrusion-type concrete printing machine to achieve the printability
  - Structural study with mechanical testing of printed concrete specimens

Research assistant (Dr. Memari) — Jun 2019 – Jan 2020

- Structural engineering aspect for cement-based material additive manufacturing
  - Study on reinforcement strategy of a concrete printing element

Graduate Research Assistant in Center for Structural Health Care Technology in Building — Jan 2012 – Jul 2015

- Post-master research assistant
- Development of real-time structural health monitoring based on average strain using low power and wireless sensors
- Development of gene-scale design, self-diagnosis, and anti-aging technology for structural systems in buildings – The National Research Laboratory Program from the Ministry of Science, ICT and Future Planning of the Republic of Korea
- Development of assessment and reduction technology of CO_2 construction phase

Graduate Research Assistant in Building Information Group Lab — Sep 2013 – Feb 2014

- Advanced architectural performance technologies based on future information and computing technologies – The Brain Korea 21 Plus Program from The National Research Foundation of Republic of Korea (09/2013 – 02/2014)

Graduate Research Assistant in Structural System and Earthquake Engineering Lab — Aug 2013 – Nov 2013

- Research on an analytical parameter of insulated precast concrete sandwich wall panels reinforced with GFRP shear grids for design process development – The Research Service for Korea Institute of Civil Engineering and Building

Technology (08/2013 – 11/2013)

Undergraduate Internship in Millennium Engineering Co. Jul 2011 – Aug 2011

- Millennium Engineering, Seoul, South Korea

Teaching Experience

Grader – CEE 6510 Structural Dynamics (Georgia Tech) 2021 Fall Semester
Grader – CE 538 Earthquake Resistant Design of Buildings (Penn State) 2019 Fall Semester
Teaching Assistant - Structural Analysis (Yonsei University) Apr 2012 – Dec 2013

Publications

Journal Articles

- (Accepted) Keunhyoung Park, Ali M. Memari, Maryam Hojati, José P. Duarte, Shadi Nazarian, Aleksandra Radlínska, Scen Bilén, "3D Concrete Printing, Material Characterization, Capacity Prediction, and Strength Testing of a Sub-scale Concrete Dome Structure", *Fronties*, 2026
- Keunhyoung Park, Ali M. Memari, Maryam Hojati, Aleksandra Radlińska, José P. Duarte, and Shadi Nazarian, "Effects of Anisotropic Mechanical Behavior on Nominal Moment Capability of 3D Printed Concrete Beam with Reinforcement", *Buildings*, 14(10), 3175, 2024. doi:10.3390/buildings14103175
- Maryam Hojati, Ali M. Memari, Mehrzad Zahabi, Zhengyu Wu, Zhanzhao Li, Keunhyoung Park, Shadi Nazarian, Jose P. Duarte, "Barbed-wire reinforcement for 3D concrete printing", *Automation in Construction*, Vol. 141, 2022. doi:10.1016/j.autcon.2022.104438
- Maryam Hojati, Zhanzhao Li, Ali M. Memari, Keunhyoung Park, Mehrzad Zahabi, Shadi Nazarian, José P. Duarte, Aleksandra Radlińska, "3D-printable quaternary cementitious materials towards sustainable development: Mixture design and mechanical properties", *Results in Engineering*, Vol. 13, 100341, 2022. doi:10.1016/j.rineng.2022.100341
- Keunhyoung Park, B.K. Oh, H.S. Park, S.W. Choi, "GA-Based Multi-Objective Optimization for Retrofit Design on a Multi-Core PC Cluster", *Computer-Aided Civil and Infrastructure Engineering*, Vol. 30, No. 12, pp. 965–980, 2015. doi:10.1111/mice.12176
- H.S. Park, Keunhyoung Park, Y.S. Kim, and S.W. Choi, "Deformation Monitoring of the Building Structure using a Motion Capture System", *IEEE/ASME Transactions on Mechatronics*, Vol. 20, No. 5, pp. 2276–2284, 2015. doi:10.1109/TMECH.2014.2374219
- Keunhyoung Park, Se Woon Choi, Y.S. Kim, H.S. Park, "A Study on the Scalability of Multi-core-PC Cluster for Seismic Design of Reinforced-Concrete Structures based on Genetic Algorithm", *Journal of Computational Structural Engineering Institute of Korea*, Vol. 26, No. 4, pp. 275–291, 2013.
- Se Woon Choi, Keunhyoung Park, Y.S. Kim, H.S. Park, "A Numerical Study on the Strain Based Monitoring Method for Lateral Structural Response of Buildings using FBG Sensors", *Journal of Computational Structural Engineering Institute of Korea*, Vol. 26, No. 4, pp. 263–269, 2013.

Conference Proceedings

- Keunhyoung Park, Ali Memari, Shadi Nazarian, J Duarte, Maryam Hojati, "Structural analysis of full-scale and sub-scale structure for digitally designed martian habitat", Proceedings of the 5th Residential Building Design and Construction Conference, 2020
- Naveen Kumar Muthumanickam, Keunhyoung Park, Jose Pinto Duarte, Shadi Nazarian, AM Memari, S Bilén, "BIM for parametric problem formulation, optioneering and 4D simulation of a 3D-printed martian habitat: A case study of the NASA 3D-printed habitat challenge", Proceedings of the 5th Residential Building Design and Construction Conference, 2020
- Keunhyoung Park, Ali Memari, Maryam Hojati, Mehrzad Zahabi, Shadi Nazarian, José Pinto Duarte, "Experimental testing and finite element modeling of 3D-printed reinforced concrete beams", Proceedings of the 5th Residential Building Design and Construction Conference, 2020
- S. W. Choi, K. H. Park, B. K. Oh, Y. S. Kim and H. S. Park, "Optimal Seismic Design Method to Induce the Beam-hinging Mechanism in Reinforced Concrete Frames", Tenth U.S. National Conference on Earthquake Engineering Frontiers of Earthquake Engineering, Anchorage, Alaska., 2014 July 21–25
- K. H. Park, S. W. Choi, Y. S. Kim and H. S. Park, "Research on the Scalability of Multi-core-PC Cluster for Seismic Structural Analysis", 2013 Computational Structural Engineering Institute of Korea, Korea
- K. H. Park, S. W. Choi, Y. S. Kim and H. S. Park, "Relationship between the Corrosion of Rebar in RC Structure with the Decline of Capacity of resistance to Earthquake", 2012 Korea institute for Structural Maintenance Inspection, Korea
- K. G. Kim, K. H. Park, S. W. Choi, Y. S. Kim and H. S. Park, "Accuracy Test of Motion Capture System Based on CCD Camera", 2012 Korea institute for Structural Maintenance Inspection, Korea
- K. H. Park, H. G. Kim, Y. A. Shin and H. S. Park, "Prediction Algorithm of Building Fragility by Considering Element Capacity Degradation of RC Structure Affected by Reinforcement Corrosion", 2012 Korea institute for Structural Maintenance Inspection, Korea

Thesis

STRUCTURAL CONSIDERATION IN ADDITIVE MANUFACTURING OF CONCRETE AS PART OF NASA'S CENTENNIAL 3D PRINTED MARS HABITAT CHALLENGE (2020)

Abstract: Additive Manufacturing (AM) method used for printing concrete offers potential improvement in conventional building construction. Numerous benefits of AM of Concrete (AMoC), such as less material waste, low labor demand, and ability to mass-customize for aesthetic purposes, allow the design flexibility combined with efficient manufacturing and lower cost for concrete construction compared to traditional methods. However, as much as AM has great potentials, there is also a lot of demand for refining AM technology for practical application to concrete construction. The development of a 3D printer system for AMoC requires comprehensive and exhaustive study covering interdisciplinary areas.

This study considered the structural engineering aspect of using AMoC for 3D printing of habitat on Mars according to guidelines specified in NASA's 3D Printed Habitat Challenge for Mars. This thesis reflects one aspect of a Penn State research team that participated in the NASA's 3D printed habitat competition, and that has been developing and deploying an AMoC system and experimenting using three different printable mixtures and operating a 3D concrete printer system for academic study and practical application. The research presented in this thesis focused on studying the structural aspects of printing a series of components as part of the habitat design development. The study helped to evaluate the possibility of 3D printed habitat's structural integrity under the Mars' surface environment. Furthermore, the study provided some lessons based on observations of printing and testing printed components. For example, the study provides some understanding related to existing anisotropy in mechanical properties, degraded structural performance in comparison to cast concrete parts, varying printability and structural performance of different printable concretes, possibility and limitation of 3D printed Mars habitat, and possible improvement of AMoC based on the challenges from the research experience.

Optimization of Seismic BRB Retrofitting Design of Structures based on Distributed Genetic Algorithm using Multi-core PC Cluster (2014)

Abstract: Structural optimization of seismic retrofitting need too much expense of computation time, because of the repetitive process for structural analysis. In this study, the optimization process took time about two to seven days for computation of performing the seismic retrofitting by using genetic algorithms in serial version. Furthermore, the successful implementation of genetic algorithm-based optimization requires a repeating investigation until to get the reasonable optimal solution. Therefore, to overcome these difficulties, a high-performance GA is developed in the form of a distributed hybrid genetic algorithm for structural optimization, implemented on a cluster of multi-core personal computers. The distributed hybrid genetic algorithm proposed in this paper consists of NSGA-II. The algorithm is implemented on a PC cluster and applied to the design of BRB retrofitting on 2-D frame steel structures and 3-D SPEAR building cases. Nonlinear static pushover analysis and dynamic analysis were performed to evaluate the seismic responses of the 2-D and 3-D structures. The fragility analysis was performed to confirm the improvement of seismic performance of retrofitted 3-D structures, SPEAR building. And repetitive optimization process was performed to get the verification of the convergence of the optimal solutions. The results show that the computation time required for GA-based optimization can be dramatically reduced and that the results of optimal solutions are stably converged. The cluster model can be employed by small scale organization as high-performance computation machine for structural optimization by low purchase cost.

Honors & Awards

Top graduate research poster Apr 2019
Research Poster, Architectural Engineering Institute

Audience favorite poster Apr 2019
Research Poster, Architectural Engineering Institute

The excellent prize Sep 2011
Graduation exhibition, Architectural Engineering department Yonsei University

Encouragement prize Jul 2011

A Structural Competitive Exhibition, The Korean Structural Engineers Association

Patent

APPARATUS AND METHOD FOR MONITORING LATERAL STRUCTURAL RESPONSE OF BUILDING BASED ON STRAIN SENSOR, Application Num. 2013-0151240, 2013-08-26

Computation Skill

Structural Analysis Programs	OpenSees, Autodesk Nastran, ABAQUS
Programming Language	MATLAB, Python